A GEOMETRICALLY CONNECTED OPEN OF A FIBRE OF A FLAT FINITELY PRESENTED MAP WITH GEOMETRICALLY REDUCED FIBRES SPREADS TO GEOMETRICALLY CONNECTED FIBRES AFTER AN ELEMENTARY ETALE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13192-13210 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13212-13255, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The proof's clause numbering mixes (a)/(b) in the statement with (1)/(2) references inside the proof; retained as the author wrote them.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The slice machinery (chapter's lemma-slice-CM, lemma-etale-makes-quasi-finite-finite-at-point, lemma-flat-finite-presentation-CM-open) and the connected-fibre descent lemma (lemma-descent-connected-fibres); plus standard inputs (finite flat implies finite locally free, fppf maps are open).

Source reference. Original lemma 057I: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13088–13103. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02LK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11924–11934. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 045U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5981–5984. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0570: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6358–6361. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 02KB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13255–13255. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01UA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13255–13255. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01K1: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13255–13255. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $h : Y \to S$ be a morphism of schemes. Let $s \in S$ be a point. Let $T \subset Y_s$ be an open subscheme. Assume*

(1) *h is flat and of finite presentation,*
(2) *all fibres of h are geometrically reduced, and*
(3) *T is geometrically connected over $\kappa(s)$.*

Then we can find an affine elementary étale neighbourhood $(S', s') \to (S, s)$ and a quasi-compact open $V \subset Y_{S'}$ such that

(a) *all fibres of $V \to S'$ are geometrically connected,*
(b) *$V_{s'} = T \times_s s'$.*

Proof. The problem is clearly local on S, hence we may replace S by an affine open neighbourhood of s. The topology on Y_s is induced from the topology on Y, see Schemes, Lemma 01K1. Hence we can find a quasi-compact open $V \subset Y$ such that $V_s = T$. The restriction of h to V is quasi-compact (as S affine and V quasi-compact), quasi-separated, locally of finite presentation, and flat hence flat of finite presentation. Thus after replacing Y by V we may assume, in addition to (1) and (2) that $Y_s = T$ and S affine.

Pick a closed point $y \in Y_s$ such that h is Cohen-Macaulay at y, see Lemma 045U. By Lemma 0570 there exists a diagram

$$\begin{array}{ccc} Z & \longrightarrow & Y \\ & \searrow & \downarrow \\ & & S \end{array}$$

such that $Z \to S$ is flat, locally of finite presentation, locally quasi-finite with $Z_s = \{y\}$. Apply Lemma 02LK to find an elementary neighbourhood $(S', s') \to (S, s)$ and an open $Z' \subset Z_{S'} = S' \times_S Z$ with $Z' \to S'$ finite with a unique point $z' \in Z'$ lying over s. Note that $Z' \to S'$ is also locally of finite presentation and flat (as an open of the base change of $Z \to S$), hence $Z' \to S'$ is finite locally free, see Morphisms, Lemma 02KB. Note that $Y_{S'} \to S'$ is flat and of finite presentation with geometrically reduced fibres as a base change of h. Also $Y_{s'} = Y_s$ is geometrically connected. Apply Lemma 057I to $Z' \to Y_{S'}$ over S' to get $V \subset Y_{S'}$ quasi-compact open satisfying (2) whose fibres over S' are either empty or geometrically connected. As $V \to S'$ is open (Morphisms, Lemma 01UA), after replacing S' by an affine open neighbourhood of s' we may assume $V \to S'$ is surjective, whence (1) holds. □